25901113

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBTG201 - BIOSTATISTICS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours Maximum Marks: 70

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

1. Data collected from websites is classified under which type? [K] / [CO1]
   Secondary data- 2 marks
2. Give the meaning of statistics. [K] / [CO1]
   Statistics is the science of collecting, organizing, analyzing, and interpreting data- 2 marks.
3. Name any two applications of biostatistics. [K] / [CO1]
   Any two applications- 2 marks
4. Give one quantitative classification example. [K] / [CO2]
   Any one (E.g., Classification based on age groups (e.g., 0–10, 10–20) - 2 marks
5. Name any one method of primary data collection. [K] / [CO3]
   Any One (Direct personal investigation, Indirect oral investigation, questionnaire/Schedule)- 2 marks
6. Name any one type of diagrammatic presentation of data. [K] / [CO3]
   Any one-E.g., Bar diagram- 2 marks
7. The level of significance is denoted as _____ [K] / [CO3]
   α (alpha)- 2 marks
8. The set of all possible outcomes of a random experiment is called_____ [U] / [CO3]
   Sample space- 2 marks
9. Define computer-oriented statistical techniques. [K] / [CO4]
   Techniques that use computers and software to perform statistical analysis, data processing, and interpretation- 2 marks
10. The spread of Normal curve depends on _____ [U] / [CO3]
    Standard deviation (σ)- 2 marks
11. What is MS Excel? [K] / [CO4]
    A spreadsheet software used for data entry, calculation, analysis, and graphical representation-2 marks

12. Name any two statistical packages. [K] / [CO4]

SPSS and R - 2 marks [2x10 = 20]

Part B-Short Essay-Answer any 4 questions

Each question carries 5 marks

13. Explain simple random sampling, stratified sampling, and systematic sampling. [U] /[CO1]

Simple random sampling: Every unit has equal chance (e.g., lottery method)

Stratified sampling: Population divided into groups (strata), samples taken from each

Systematic sampling: Select every kth unit (e.g., every 5th person)

Explanation – 4 marks

Example- 1 mark

14. Compare qualitative data and quantitative data with clear distinguishing points. [U] / [CO1]

Compare qualitative data -2.5 marks, and quantitative data - 2.5 marks):

Qualitative: Non-numerical (e.g., gender, color)

Quantitative: Numerical (e.g., height, weight)

Qualitative → categories

Quantitative → measurable values

15. What are the characteristics of classification? [K] / [CO2]

Unambiguity, Simplicity, Flexibility, Suitability, Homogeneity, Exhaustiveness

6 characteristics – 5 marks

16. Explain the concept of central tendency. [K] / [CO2, CO3]

It refers to a single value representing the entire dataset.

Main measures: Mean, Median, Mode

General characteristics: 2 marks, Mean, Median and Mode: 1 mark each (3 marks)

17. Briefly explain ANOVA. [U] / [CO3]

A statistical method used to compare means of more than two groups to see if there is a

Significant difference.-Introduction – 2 marks, Steps with f ratio 2 marks, Uses – 1 mark

18. Explain computation of mean. [A] / [CO4]

Arithmetic mean = $\Sigma X / N$, Where ΣX = sum of observations, N = number of observations

Calculation – 3 marks, E.g. - 2 marks [5x4 = 20]

Part C-Essay type questions-Answer any 2 questions

Each question carries 15 marks

19. Describe various measurers of deviation with their merits and limitations. [K] / [CO2, CO3]

Range, Quartile Deviation, Mean deviation, Variance, Standard deviation-Description (3 marks), calculation (6 marks) Merits (3 marks) and demerits (3 marks) of all the methods

20. Explain in detail the concept and types correlation. [U] / [CO2, CO3]

Correlation – concepts- Measures relationship between variables-(4 marks)

Types: (6 marks)

Positive, Negative, Zero

Simple, Multiple, Partial

Methods: Scatter diagram, Karl Pearson coefficient (5 marks)

21. Explain Pie diagram. Represent the following data of pupulations of different mammals in a reserve forest in Tamil Nadu using a pie diagram.

Deers - 105 , Monkeys - 65 , Fox - 50 , Leopards - 35 , Tigers - 20 ,Others - 25. [A] / [CO3]

Circular chart showing proportions-Total = 360°,

Given data total = 105+65+50+35+20+25 = 300

Angles: • Deer = (105/300) ×360 = 126° • Monkeys = 78° • Fox = 60°• Leopards = 42° • Tigers = 24° • Others = 30°

Calculation – 8 marks, Graph – 7 marks

22. Explain Bubble sort in detail with example. [A] / [CO4]

Bubble sort-A sorting algorithm, repeatedly compares adjacent elements and swaps if needed

Example:

Input: 5, 3, 2, 4

Step 1: 3, 2, 4, 5

Step 2: 2, 3, 4, 5 (sorted)

Explanation – 10 marks

Example.- 5 marks [15x2 = 30]